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Universal Basic Income: A Dynamic Assessment†

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Universal basic income (UBI) is an increasingly popular policy proposal, but there is no evidence regarding its longer-term consequences. We find that UBI generates large welfare losses in a general equilibrium model with imperfect capital markets, labor market shocks, and intergenerational linkages via skill formation and transfers. This conclusion is robust to various alternative ways of financing UBI. By using observationally equivalent models that eliminate different sources of endogenous dynamic linkages (equilibrium capital market and parental investment in child skills), we show that the latter are largely responsible for the negative welfare consequences. (*JEL* H24, H31, I38, J22, J24, J62)

The idea of universal basic income (UBI)—a set income that is given to all without any conditions—is making an important comeback in many countries. This is most likely the result of the large increase in inequality over the last few decades and stagnant median wages, as well as anxieties about automation and robotization and, more recently, the increase in unemployment in response to the coronavirus pandemic and uncertainty about future prospects.[1] What do we know about UBIs?[2] What problems would a UBI ameliorate, and which would it aggravate?

There is no real experience in advanced economies with a UBI policy, although studies have made use of variation in income arising from changes in oil revenue

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[1] A UBI policy has been advocated by people ranging from Pope Francis, to Elon Musk, to former US presidential candidate Andrew Yang, as well as by senior officials in organizations such as the United Nations or the World Economic Forum. See Wignaraja (2020).

[2] See Gentilini et al. (2019) for a recent excellent review of UBI history and lessons from diverse country experiences.

or Earned Income Tax Credit (EITC) generosity to study potential consequences.[3] Although much of the attention has been on the effects of programs on labor supply, it may very well be that the more important consequences of a UBI are long run. There have not been, however, long-run, large-scale experiments that allow one to evaluate the longer-term consequences of these programs. As stated by Hoynes and Rothstein (2019, p. 952) in their excellent review article on UBI in advanced economies,

> we have a good deal of evidence from a range of settings that substitution effects on short-run labor supply are moderate and income effects are small. There is also clear evidence that additional family resources improve children's outcomes, including health and school achievement. The major open questions about UBIs, in our view, relate to longer-run effects, which are much harder to study using randomized and natural experiments.[4]

How may the longer-run effects of UBI differ from those in the short run? What are the consequences (from changes in prices and taxes) of a large-scale UBI? There are various dimensions in which the policy consequences of a long-run, large-scale program may differ from those of a small-scale or shorter-run policy. First, and more traditionally, studying shorter-term or more local experiments will not capture the longer-run general equilibrium consequences of such a policy on variables ranging from the capital stock to wages and tax rates. Second, it may very well be that the more important consequences of a UBI are intergenerational. For example, policies that increased maternal employment and family income were found to increase child achievement (Morris et al. 2009). Programs such as SNAP and the EITC have been shown to improve health at birth (e.g., Almond, Hoynes, and Schanzenbach 2011), and increased generosity in the EITC is also associated with higher children's achievement (Dahl and Lochner 2012; Chetty, Friedman, and Rockoff 2011) and educational attainment (Bastian and Michelmore 2018; Manoli and Turner 2018). The universality and presumably generous payments implied by UBI may mean very different longer-run intergenerational consequences, however.

This paper provides a very inexpensive evaluation of such a UBI policy by studying its consequences in a computational model laboratory. We develop a model that incorporates many of the most important channels that affect the costs and benefits associated with a UBI policy. The model features an economy with imperfect capital markets and overlapping generations. An individual's first decision is an education choice (college) based on their assets, skills, and their taste for education. Skills and initial assets are endogenous: the first is the result of investments made by parents during an individual's childhood, and the second is a parental transfer. College can be financed with a combination of the parental transfer, working while in college, and borrowing. After education, an individual works, has children, invests in their skills, makes a monetary transfer to their child, and eventually retires and dies. These intergenerational linkages are embedded in a fairly standard general equilibrium

[3] The Alaska Permanent Fund and the Eastern Band of Cherokee Indians tribal government provide demogrants to adults. The first makes payments that may vary from year to year, ranging from $1,000–$2,000 per person per year and financed by Alaska's oil revenues. The second provides payments of around $4,000 per person per year financed out of tribal casino revenues. See Jones and Marinescu (2022); Akee et al. (2010); and Akee et al. (2018).

[4] Emphasis is ours.

life cycle Aiyagari framework with wage uncertainty and with a tax function calibrated to the US economy. This framework allows aggregate education, skills, labor supply, and savings to affect prices, and the endogeneity of these outcomes means that they are affected by the additional income provided by UBI and via the change in taxes required to finance this policy.

The steady state of the model is parameterized and estimated to match household-level data near the year 2000 (the base year for prices) using a variety of data sources, such as the Panel Study of Income Dynamics (PSID) and the 1979 cohort of the National Longitudinal Survey of Youth (NLSY79). Importantly, given Heathcote and Tsujiyama (2021) finding that the space for further redistribution depends on how large government expenditures are with respect to national income, we calibrate the model to match the appropriate ratio for the US economy. We validate the model in a variety of ways that are economically significant for the question we are studying. In particular, we show that the model does a good job of matching labor income and wealth inequality by quintile (all nontargeted moments) as well as measures of intergenerational mobility. It also matches various nontargeted measures of inequality of investment in children, both in their skills and in monetary transfers by their parents, as well as statistics regarding student loans. Given the importance of labor supply responses and the (less well-studied) response of parental investment in children's skills to income/wealth shocks, we conduct the appropriate partial equilibrium exercises in the model to compare its predictions with those of the effects of lottery winnings on labor earnings (Golosov et al. 2021), the effect of nonlabor income on labor supply (Blundell and MaCurdy 1999), and the impact of cash transfers on child development (Dahl and Lochner 2012). We show that the model generates responses in line with the empirical literature.

We introduce the UBI policy as a lump-sum transfer that unconditionally gives all adults a yearly income equivalent of $8,000 as measured in year 2000 dollars.[5] What are the potential benefits of a UBI policy? In an economy in which individuals are subject to both wage and employment shocks and in which credit and insurance markets are imperfect, UBI allows for greater smoothing of consumption and the guarantee of a minimum standard of living. It can also allow agents to undertake relatively expensive investments (in our model, attend college) at a lower cost than via borrowing. Furthermore, it can have beneficial intergenerational consequences by facilitating parental investment in their child's skill formation and transferring resources to them when they become adults.[6] Of course, any positive effects of UBI must be weighed against the cost of increased distortionary taxation and potentially negative general equilibrium effects in order to assess the net welfare impact.

Using a utilitarian welfare function, we find that a UBI policy, despite decreasing inequality, has large negative welfare consequences for cohorts that are adults when the policy is introduced (6 percent in consumption equivalence units) and

[5] This is equivalent to $1,000 per month in 2020 dollars, as proposed, for example, in Andrew Yang's 2020 presidential campaign. It is also the amount evaluated by Hoynes and Rothstein (2019) and Golosov et al. (2021), among others.

[6] The literature on UBI has also pointed to other, mainly psychological and mental health benefits, which we do not assess here but are important to consider as well in a fuller evaluation of this policy. There are also important ethical and philosophical arguments made on its behalf (Van Parijs and Vanderborght 2017). See also Ghatak and Maniquet (2019) for a theoretical assessment of the desirability of a UBI.

significantly larger negative welfare implications for subsequent cohorts and in the new steady state (22 percent). Labor supplied falls immediately, and the capital stock decreases dramatically over time. Intergenerationally, the higher labor tax reduces investment in children's skills, decreases parental transfers, and lowers the share of agents with college education. In aggregate terms, we find that the UBI policy is associated with a long-run GDP reduction of almost 20 percent, of which the fall in the capital stock explains half, with the decline in labor supply and in efficiency units of labor being responsible for the remainder.

An important contribution of our paper is its investigation of the main channels responsible for the negative welfare results. We conduct several exercises that illuminate the dynamic roles of capital and intergenerational linkages. Our first exercise shows that the negative effects of UBI are much greater when UBI is received by all rather than given to only one cohort, even in a partial equilibrium environment. In fact, general equilibrium effects lessen the negative consequences of the policy. Second, the OLG structure of the economy allows us to conduct a novel analysis in which we study adjacent cohorts that differ only in whether UBI was introduced before versus after their parents had invested in their skill formation and before versus after their monetary transfer. These comparisons allow us to show that the first cohort whose state variables are all determined under UBI already suffers 57 percent of the steady-state losses. Furthermore, comparing those cohorts that differ only in whether they received skill investment versus skills and parental transfers prior to UBI, we show that the decline in skills is associated with a larger proportion of the welfare loss. Third, assessing welfare in the new steady state, we show that even if all value functions were kept at the same level as in the original steady state of the non-UBI benchmark economy, the change in distribution over these states is responsible for 47 percent of the welfare loss. As the distribution over states is determined by parents via their investment in child skills and their monetary transfers, this illustrates the importance of these linkages, which are absent in models in which each generation starts de novo.

Turning next to the main mechanisms, we show that if we replace endogenous skills with an exogenous (and, therefore, policy invariant) transition matrix from parental education and skill to child skills, the long-run welfare losses are halved, but the welfare losses of adults when the policy is introduced are relatively unaffected. The steady-state welfare loss is even smaller, and adults actually gain, in a different experiment in which the aggregate capital stock and interest rate are kept constant although agents are still allowed to borrow and save (at the same steady-state interest rate). This modification of the model reduces the amount of labor taxation required to fund UBI, as revenue from capital taxation remains unchanged, which in turn allows for a smaller reduction in labor supply. Thus, the endogenous accumulation of human and physical capital (with its equilibrium effects on tax revenues) is the key driver of the negative welfare results.

The Mirrleesian public finance literature has shown the usefulness of examining small changes in the tax and transfer system and decomposing their effect into mechanical, behavioral, and general equilibrium components.[7] Conducting the same

[7] We thank the editor for suggesting that we follow this approach.